Problem 3

Part a)

Calculation of Marginal Distributions: We first determine the marginal densities $f_X(x)$ and $f_Y(y)$ using the joint density.

$$\begin{aligned}
f_X(x) &= \int_{\mathbb{R}} f_{(X,Y)}(x,y)\,dy \\
&= \int_{\mathbb{R}} \frac{1}{4}(1+xy)\mathbf{1}_{[-1,1]^2}(x,y)\,dy \\
&= \int_{\mathbb{R}} \frac{1}{4}(1+xy)\mathbf{1}_{[-1,1]}(x)\mathbf{1}_{[-1,1]}(y)\,dy \\
&= \frac{1}{4}\mathbf{1}_{[-1,1]}(x)\int_{-1}^{1}(1+xy)\,dy \\
&= \frac{1}{4}\mathbf{1}_{[-1,1]}(x)\left[y + x\frac{y^2}{2}\right]_{-1}^{1} \\
&= \frac{1}{4}\mathbf{1}_{[-1,1]}(x)\left(\left(1+\frac{x}{2}\right) - \left(-1+\frac{x}{2}\right)\right) \\
&= \frac{1}{4}(2)\mathbf{1}_{[-1,1]}(x) \\
&= \frac{1}{2}\mathbf{1}_{[-1,1]}(x)
\end{aligned}$$

By symmetry of the joint density function, the marginal density of Y is identical:

$$f_Y(y) = \frac{1}{2}\mathbf{1}_{[-1,1]}(y)$$

Thus, X and Y are both uniformly distributed on $[-1,1]$, i.e., $X, Y \sim \mathcal{U}(-1,1)$.

Part a) i)

- **Calculation of $\mathbb{E}\left[X\mathbf{1}_{\{X<1/2\}}\right]$:** Since $|X\mathbf{1}_{\{X<1/2\}}| \le 1$, the random variable is bounded and thus its expectation exists. Using the marginal density $f_X(x) = \frac{1}{2}\mathbf{1}_{[-1,1]}(x)$ and Corollary 3.3.13.:

$$\begin{aligned}
\mathbb{E}\left[X\mathbf{1}_{\{X<1/2\}}\right] &= \int_{\mathbb{R}} x\mathbf{1}_{(-\infty,1/2)}(x)f_X(x)\,dx \\
&= \int_{\mathbb{R}} x\mathbf{1}_{(-\infty,1/2)}(x)\frac{1}{2}\mathbf{1}_{[-1,1]}(x)\,dx \\
&= \int_{\mathbb{R}} \frac{x}{2}\mathbf{1}_{[-1,\frac{1}{2})}(x)\,dx \\
&= \int_{-1}^{1/2}\frac{x}{2}\,dx \\
&= \left[\frac{x^2}{4}\right]_{-1}^{1/2} = \left(\frac{1}{16}-\frac{1}{4}\right) = \frac{1}{16}-\frac{4}{16} = -\frac{3}{16}
\end{aligned}$$

- **Calculation of $\mathbb{E}\left[\frac{1}{X}\right]$:** Using the marginal density $f_X(x) = \frac{1}{2}\mathbf{1}_{[-1,1]}(x)$ and the substitution formula (Corollary 3.3.13), the expectation is given by the integral:

$$\mathbb{E}\left[\frac{1}{X}\right] = \int_{\mathbb{R}} \frac{1}{x}f_X(x)\,dx = \int_{-1}^{1}\frac{1}{2x}\,dx$$

This is an improper integral due to the singularity at $x = 0$. Let $Z = \frac{1}{X}$. According to Definition 3.3.7, for the expected value to exist, we must check the integrability of the positive part $Z^+ = \max(0, Z)$ and the negative part $Z^- = -\min(0, Z)$.

Evaluating the expectation of the positive part:

$$\begin{aligned}
\mathbb{E}[Z^+] &= \int_0^1 \frac{1}{2x}\,dx \\
&= \lim_{\epsilon \searrow 0} \left[\frac{1}{2}\ln(x)\right]_\epsilon^1 \\
&= \lim_{\epsilon \searrow 0} \left(0 - \frac{1}{2}\ln(\epsilon)\right) = \infty
\end{aligned}$$

Evaluating the expectation of the negative part:

$$\begin{aligned}
\mathbb{E}[Z^-] &= \int_{-1}^0 \left(-\frac{1}{2x}\right) dx \\
&= \lim_{\epsilon \nearrow 0} \left[-\frac{1}{2}\ln|x|\right]_{-1}^\epsilon \\
&= \lim_{\epsilon \nearrow 0} \left(-\frac{1}{2}\ln|\epsilon| - 0\right) = \infty
\end{aligned}$$

Since $\mathbb{E}[Z^+] = \mathbb{E}[Z^-] = \infty$, the expected value $\mathbb{E}\left[\frac{1}{X}\right]$ **does not exist** (Definition 3.3.7(iii)).

- **Calculation of $\mathbb{E}[XY]$:**

  Using the joint density and the substitution formula in $\mathbb{R}^2$ (Theorem 3.3.12):

$$\begin{aligned}
\mathbb{E}[XY] &= \int_{\mathbb{R}^2} xy \cdot f_{(X,Y)}(x,y)\,d\lambda^2(x,y) \\
&= \int_{\mathbb{R}^2} xy \cdot \frac{1}{4}(1+xy)\mathbf{1}_{[-1,1]^2}(x,y)\,d\lambda^2(x,y)
\end{aligned}$$

  Since $|xy| \le 1$ on the support $[-1,1]^2$, we have $\mathbb{E}[|XY|] \le 1 < \infty$. Thus, Fubini's theorem applies and we can arbitrarily choose the order of integration:

$$\begin{aligned}
&= \int_{\mathbb{R}}\int_{\mathbb{R}} xy \cdot \frac{1}{4}(1+xy)\mathbf{1}_{[-1,1]^2}(x,y)\,dy\,dx \\
&= \frac{1}{4}\int_{-1}^1\int_{-1}^1 (xy + x^2y^2)\,dy\,dx \\
&= \frac{1}{4}\int_{-1}^1 x\left[\frac{y^2}{2}\right]_{-1}^1 + x^2\left[\frac{y^3}{3}\right]_{-1}^1\,dx \\
&= \frac{1}{4}\int_{-1}^1 \left(x\left(\frac{1}{2} - \frac{1}{2}\right) + x^2\left(\frac{1}{3} - \left(-\frac{1}{3}\right)\right)\right) dx \\
&= \frac{1}{4}\int_{-1}^1 \left(0 + x^2\left(\frac{2}{3}\right)\right) dx \\
&= \frac{1}{4}\int_{-1}^1 \frac{2x^2}{3}\,dx = \frac{1}{6}\int_{-1}^1 x^2\,dx \\
&= \frac{1}{6}\left[\frac{x^3}{3}\right]_{-1}^1 = \frac{1}{6}\left(\frac{1}{3} - \left(-\frac{1}{3}\right)\right) = \frac{1}{6}\left(\frac{2}{3}\right) = \frac{2}{18} = \frac{1}{9}
\end{aligned}$$

Part a) ii)

According to Theorem 3.2.3 c), the continuous random variables X and Y are independent if and only if $f_{(X,Y)}(x,y) = f_X(x)f_Y(y)$ for Lebesgue-almost all $(x,y) \in \mathbb{R}^2$. The product of the marginals is:

$$f_X(x)f_Y(y) = \frac{1}{4}\mathbf{1}_{[-1,1]^2}(x,y).$$

Since $\frac{1}{4}\mathbf{1}_{[-1,1]^2}(x,y) \neq \frac{1}{4}(1+xy)\mathbf{1}_{[-1,1]^2}(x,y)$ on a set of non-zero Lebesgue measure (specifically, the regions where $xy \neq 0$), X and Y are not independent.

Part b)

We apply the Law of Total Probability (Theorem 2.1.7) to the countable partition $\{Y = k\}_{k\in\mathbb{N}_0}$:

$$\begin{aligned}\mathbb{P}(X > Y) &= \sum_{k=0}^{\infty} \mathbb{P}(X > Y \mid Y = k)\mathbb{P}(Y = k) \\ &= \sum_{k=0}^{\infty} \mathbb{P}(X > k \mid Y = k)\mathbb{P}(Y = k). \qquad (1)\end{aligned}$$

By Definition 3.2.1, X and Y are independent, meaning $\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$. We use the definition of conditional probability to simplify the conditional term:

$$\mathbb{P}(X > k \mid Y = k) = \frac{\mathbb{P}(X > k, Y = k)}{\mathbb{P}(Y = k)} = \frac{\mathbb{P}(X > k)\mathbb{P}(Y = k)}{\mathbb{P}(Y = k)} = \mathbb{P}(X > k).$$

Substituting this back into (1) and using the probability mass function for $Y \sim \text{Geom}(p)$ gives:

$$\mathbb{P}(X > Y) = \sum_{k=0}^{\infty} \mathbb{P}(X > k)p(1-p)^k. \qquad (2)$$

To evaluate $\mathbb{P}(X > k)$, we integrate the density of $X \sim \text{Exp}(\lambda)$. Because the outer sum iterates over $Y \sim \text{Geom}(p)$ where $k \in \mathbb{N}_0$, we know $k \geq 0$. Thus, the indicator function is 1 over the entire interval $[k, \infty)$:

$$\mathbb{P}(X > k) = \int_k^{\infty} \lambda e^{-\lambda x}\mathbf{1}_{[0,\infty)}(x)\,dx = \int_k^{\infty} \lambda e^{-\lambda x}\,dx.$$

Substituting $u = -\lambda x$, so $du = -\lambda\,dx$, with bounds $x = k \Rightarrow u = -\lambda k$ and $x \to \infty \Rightarrow u \to -\infty$:

$$\mathbb{P}(X > k) = \int_{-\lambda k}^{-\infty} (-1)e^u\,du = \int_{-\infty}^{-\lambda k} e^u\,du = [e^u]_{-\infty}^{-\lambda k} = e^{-\lambda k} - \lim_{a\to-\infty} e^a = e^{-\lambda k}.$$

Substituting this result back into (2) yields:

$$\begin{aligned}\mathbb{P}(X > Y) &= \sum_{k=0}^{\infty} e^{-\lambda k} \cdot p(1-p)^k \\ &= p\sum_{k=0}^{\infty} \left(e^{-\lambda}(1-p)\right)^k.\end{aligned}$$

Since $\lambda > 0$ implies $0 < e^{-\lambda} < 1$ and $p \in (0,1)$ implies $0 < (1-p) < 1$, we have $q := e^{-\lambda}(1-p) \in (0,1)$, so we can apply the geometric series formula $\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$:

$$\begin{aligned}\mathbb{P}(X > Y) &= p\left(\frac{1}{1 - e^{-\lambda}(1-p)}\right) \\ &= \frac{p}{1 - e^{-\lambda}(1-p)}.\end{aligned}$$